

Lecture 1: Basic Concepts

EE201 – Circuit Theory I

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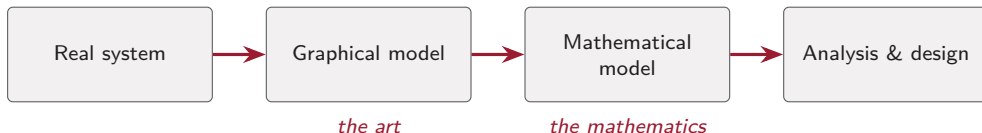
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charge  current  voltage  power 

these colours mean the same thing on every slide and in the notes

Circuit theory is not about wires



- A real resistor has inductance, a temperature coefficient, a size, a price.
- We keep **one number**, R , and throw the rest away.
- Whether that is brilliant or foolish depends on the question you are asking.

“All models are wrong, but some are useful.”

George E. P. Box, 1976

The same pipeline, other domains

Domain	“Across” quantity	“Through” quantity
electrical	electric potential	current
hydraulic	pressure	volumetric flow
thermal	temperature	heat flow
mechanical	velocity	force

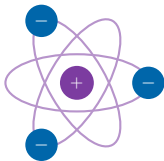
Every “across” quantity is a difference

It is always measured between two points — and in every domain the word “difference” is usually dropped. We say **voltage**, not “potential difference”; a plumber says “pressure”, not “pressure difference”. The difference is still there, hidden in the name.

Charge

Definition

Charge is the property of matter that makes it feel a force in an electromagnetic field. Symbol q , unit **coulomb (C)**.



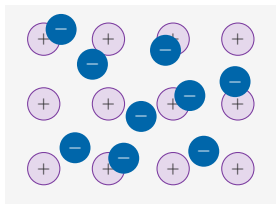
one atom: as many $-$ as $+$

$$e = 1.602 \times 10^{-19} \text{ C}$$

$$1 \text{ C} \approx 6.24 \times 10^{18} \text{ electron charges}$$

- electron: $-e$ proton: $+e$
- **Charge is conserved** — never created or destroyed, only moved. (This will become KCL.)

A wire is already full of charge



copper: fixed + ions,
electrons free to wander

1 cm³ of copper contains about

8.5×10^{22} free electrons $\approx$ 13 600 C

of *mobile* charge — and its net charge is **zero**.

So what is a current?

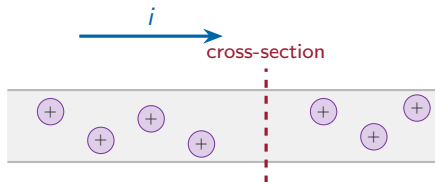
Every free electron left a + ion behind. The metal stays neutral; current is that charge *drifting*, not charge being poured in.

Current: charge in motion

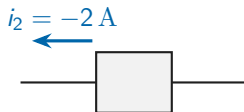
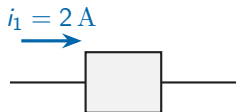
Definition

$$i(t) = \frac{dq(t)}{dt} \quad \Longleftrightarrow \quad q(t) = q(t_0) + \int_{t_0}^t i(\tau) d\tau$$

Unit: **ampere (A)**, $1 \text{ A} = 1 \text{ C/s}$.



The arrow is a choice, not a fact



Same physical situation. The arrow says which way we call positive; the sign of the number says which way the charge really goes.

A negative answer is not a mistake — it is the model telling you something.

Two numbers worth knowing

How much is a coulomb?

A 4000 mAh phone battery:

$$Q = 4 \text{ A} \times 3600 \text{ s} = 14\,400 \text{ C}$$

$$\approx 9 \times 10^{22} \text{ elementary charges}$$

At 10^{22} charges, the lumpiness of charge is invisible. We may treat $q(t)$ as a **smooth** quantity — which is what lets us write dq/dt at all.

How fast do the charges move?

1 A in a 1 mm^2 copper wire:

each electron drifts at about

$$0.07 \text{ mm/s} \approx 1 \text{ metre in 4 hours}$$

Yet the lamp lights the instant you flip the switch. The carriers crawl; the *signal* that starts them travels near the speed of light.

We come back to that gap when we define a *lumped* circuit.

Two footnotes that matter

Conventional current

The arrow points the way *positive* charge would move. In a copper wire the carriers are electrons, so they actually move *against* the arrow.

This changes no equation in this course: a current only counts the **net rate at which charge crosses a section**, and one electron moving left counts exactly the same as one positive charge moving right.

Notation

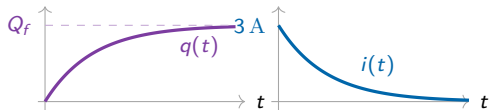
$i(t)$, $v(t)$, $q(t)$, $p(t)$ (may vary in time) I , V , Q , P (constant)



Example: charging a phone

The last part of a charge cycle: the charge *stored in the battery* approaches its full value, $q(t) = Q_f (1 - e^{-t/\tau})$, with $Q_f = 3600 \text{ C}$ and $\tau = 20 \text{ min}$.

$$i(t) = \frac{dq}{dt} = \frac{Q_f}{\tau} e^{-t/\tau}$$

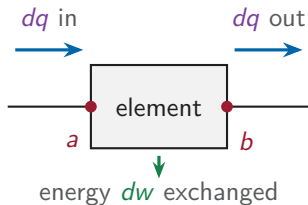


- $i(0) = 3600/1200 = 3 \text{ A}$
- after 20 min: $i = 3/e \approx 1.1 \text{ A}$
- the fuller it gets, the slower it fills

Charge is what accumulates; current is how fast it accumulates.

Voltage: why the charge moves

Charge does not move by itself. Pushing it through an element **costs or releases energy** — and that is what voltage measures.



Definition

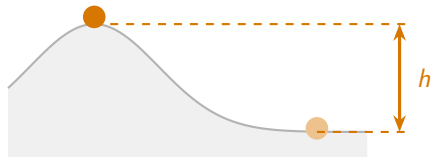
$$v_{ab} = \frac{dw}{dq} \quad \text{unit: } \mathbf{volt\ (V)}, \quad 1\text{ V} = 1\text{ J/C}$$

the energy exchanged *per unit of charge* carried from *a* to *b* through the element.

What that definition does — and does not — say

- It is a **ratio**. Carry twice the charge, exchange twice the energy: the ratio is the same. So voltage belongs to the **pair of points**, not to the charge you used to probe it.
- It is defined even when **nothing is moving**. A 1.5 V battery on the shelf has 1.5 V across it with $i = 0$ — just as a closed tap still has pressure behind it.
- Order matters: $v_{ab} = -v_{ba}$. There is no “voltage at a point” until we fix a reference (Section: ground).
- In a *lumped* circuit the energy per charge depends only on the two endpoints, **not on the path** taken between them. That is what makes “the voltage between a and b ” a meaningful phrase — and it becomes KVL.

Voltage is an electrical height

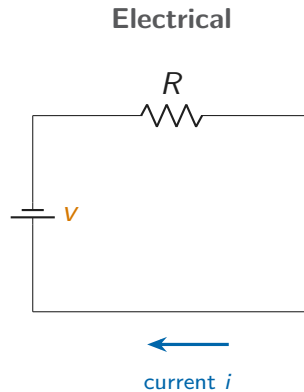
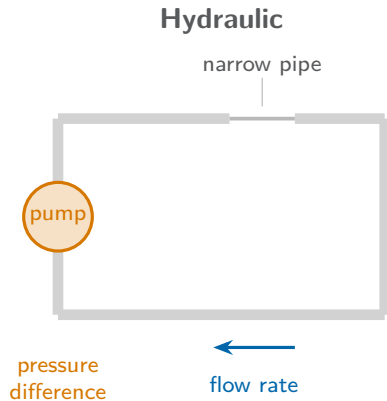


energy per unit *mass* to lift: $g h$

- Only the height **difference** matters.
- “Sea level” is a **choice** — and changing it changes no physics.
- A heavier ball gains more energy, but energy *per kilogram* is the same: that is why gh , like voltage, is a property of the two places.

Voltage is the electrical version of this, with charge in place of mass.

The hydraulic analogy



Across and through

Hydraulic	Electrical	
pressure (difference)	voltage v	across: a difference between two points
volumetric flow rate	current i	through: needs a direction
pipe restriction	resistance R	opposes flow, dissipates energy
pump	voltage source	supplies energy

Where the analogy breaks — and a warning

Water can leak out and pile up in a bucket; charge in a lumped circuit cannot. And “the current across the resistor” or “the voltage through the resistor” is not a slip of the tongue — it is a conceptual error.

Ground: the node we choose to call zero

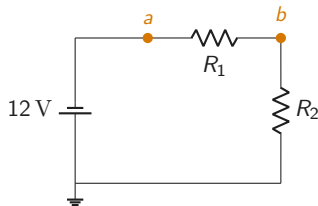
Reference node

Only differences matter, so we are free to declare one node to be 0 V. Then

$$V_{ab} = V_a - V_b.$$

Move the reference and *every* node voltage shifts by the same constant. No two-point voltage, current or power changes.

Pick the node that the most elements touch — your equations get shorter.



Three different things are called “ground”

Earth ground

A physical connection to the planet. Safety.

Chassis ground

The metal enclosure of the instrument.

Signal ground

The analysis reference. **This is ours.**

In EE201 we always mean the third one.

In the laboratory, confusing them is how you blow a fuse.

Power falls out of the definitions

$$p(t) = \frac{dw}{dt} = \underbrace{\frac{dw}{dq}}_{v(t)} \cdot \underbrace{\frac{dq}{dt}}_{i(t)} = v(t) i(t)$$

Power

unit **watt (W)**, $1 \text{ W} = 1 \text{ J/s}$

Energy

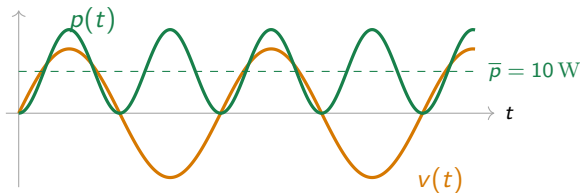
$$w = \int_{t_0}^t v(\tau) i(\tau) d\tau$$

$p = vi$ was **not postulated** — it is a consequence of defining v as energy per charge and i as charge per time.

Power that changes in time

A resistive element with $v(t) = 10 \cos \omega t$ V and $i(t) = 2 \cos \omega t$ A:

$$p(t) = vi = 20 \cos^2 \omega t = 10(1 + \cos 2\omega t) \text{ W}$$



- $p(t) \geq 0$: a resistor never gives energy back.
- The power ripples at **twice** the frequency, around $\bar{p} = 10 \text{ W}$.

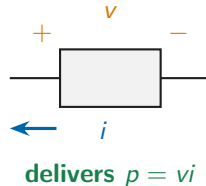
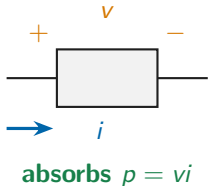
The passive sign convention

PSC

Draw the current arrow so that it **enters the + terminal**. Then

$$p = vi = \text{power absorbed by the element.}$$

$p > 0$: the element absorbs. $p < 0$: it delivers.



When the picture does not obey the PSC

Do not memorise a second formula.

Flip one reference and change that variable's sign

$$p_{\text{absorbed}} = v(-i) = -vi$$

One convention, applied consistently,
beats four memorised cases.

Conservation of power

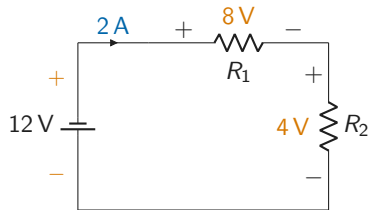
In any circuit, at every instant

$$\sum_{k=1}^N p_k(t) = 0$$

Sources ($p_k < 0$) supply exactly what the rest ($p_k > 0$) consume.

Use it. If the powers in your solution do not cancel, your solution is wrong — and you found out without looking at the answer key.

Example: a power balance



$$p_{R_1} = 8 \times 2 = +16 \text{ W}$$

$$p_{R_2} = 4 \times 2 = +8 \text{ W}$$

$$p_{\text{src}} = 12 \times (-2) = -24 \text{ W}$$

$$16 + 8 - 24 = 0 \checkmark$$

The source current *leaves* its + terminal —
hence -2 A .

What exactly is a circuit?

Electric circuit

An interconnection of electrical elements through which charge can flow along closed paths.

The lumped model — the assumption behind all of EE201

- 1 no net charge accumulates inside an element;
- 2 no magnetic flux links any region outside the elements;
- 3 the circuit is small compared with the wavelength: $d \ll \lambda = c/f$.

Then: **ordinary** differential equations in time. Otherwise: Maxwell's equations. A different course.

Lumped or not? It depends on the signal

LUMPED

mains, $f = 50$ Hz

$\lambda = 6000$ km

a 30 cm board: $d/\lambda \sim 10^{-7}$

NOT LUMPED

Wi-Fi, $f = 2.4$ GHz

$\lambda = 12.5$ cm

a 3 cm trace: $d/\lambda \sim 1/4$

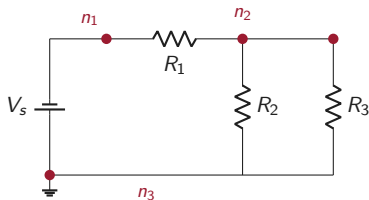
A phone motherboard is a lumped circuit for its battery
and a transmission line for its antenna.

The model is chosen by the question, not by the object.

Terminals, nodes, branches, loops

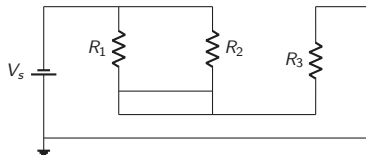
- **Terminal** — a point where an element connects. R, L, C: two. Transistor: three.
- **Node** — a connection point. Everything joined by ideal wire is **one** node, however long the wire.
- **Branch** — one element between two nodes.
- **Loop** — a closed path through no node twice.

$$\# \text{branches} = \# \text{loops} + \# \text{nodes} - 1$$

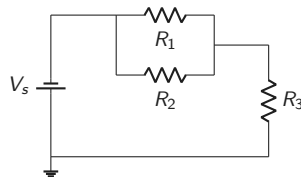


$$4 = 2 + 3 - 1$$

The same circuit, drawn twice



drawing (a)



drawing (b)

Same circuit. Drawing is not topology.

When a problem looks impossible, redraw it with the nodes lined up.

Summary

Quantity	Symbol	Definition	Unit
charge	q	—	C
current	i	$i = dq/dt$	A = C/s
voltage	v	$v = dw/dq$	V = J/C
power	p	$p = vi$	W = J/s
energy	w	$w = \int p dt$	J

- References (arrow, polarity) are **choices**; signs carry the physics.
- PSC: current into $+$ $\Rightarrow p = vi$ is absorbed, and $\sum_k p_k = 0$.
- Across vs through. Ground is a choice.
- Lumped $\Rightarrow$ ODEs. That is what makes this course possible.

Next lecture

Interconnection laws

Kirchhoff's current law (conservation of charge) and Kirchhoff's voltage law (conservation of energy), then the branch relations of the basic elements.

Questions?

Notes for this lecture: see the Section 2 repository.